

P-12

**Academic Session 2024 – 25****PHYSICS Code – 042****CLASS – XII****Maximum Marks: 70****Time Allowed: 3 hours**

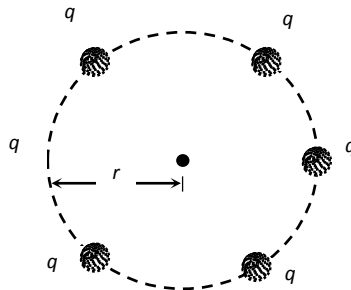
General Instructions:

- (1) There are 33 questions in all. All questions are compulsory.
- (2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.
- (3) All the sections are compulsory.
- (4) Section A contains sixteen questions, twelve MCQ and four Assertion Reasoning based of 1 mark each, Section B contains five questions of two marks each, Section C contains seven questions of three marks each, Section D contains two case study-based questions of four marks each and Section E contains three long answer questions of five marks each.
- (5) There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C, one question in each CBQ in Section D and all three questions in Section E. You have to attempt only one of the choices in such questions.
- (6) Use of calculators is not allowed.
- (7) You may use the following values of physical constants where ever necessary:
  - i.  $c = 3 \times 10^8 \text{ m/s}$
  - ii.  $m_e = 9.1 \times 10^{-31} \text{ kg}$
  - iii.  $m_p = 1.7 \times 10^{-27} \text{ kg}$
  - iv.  $e = 1.6 \times 10^{-19} \text{ C}$
  - v.  $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$
  - vi.  $h = 6.63 \times 10^{-34} \text{ J s}$
  - vii.  $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$
  - viii. *Avogadro's number*  $= 6.023 \times 10^{23} \text{ per gram mole.}$

## [SECTION – A]

(16x1=16 marks)

Q1. A point charge is surrounded symmetrically by six identical charges at distance  $r$  as shown in the figure. How much work is done by the forces of electrostatic repulsion when the point charge  $q$  at the centre is removed at infinity



(a) Zero

(b)  $\frac{6q^2}{4\pi\epsilon_0 r}$

(c)  $\frac{q^2}{4\pi\epsilon_0 r}$

(d)  $\frac{12q^2}{4\pi\epsilon_0 r}$

Q2. There are two metallic spheres of same radii but one is solid and the other is hollow, then

- (a) Solid sphere can be given more charge.
- (b) Hollow sphere can be given more charge.
- (c) They can be charged equally (maximum).
- (d) None of the above.

Q3. Which of the following has a negative temperature coefficient

- (a) C
- (b) Fe
- (c) Mn
- (d) Ag.

Q4. The magnetic field at the centre of current carrying coil is

- (a)  $\frac{\mu_0 ni}{2r}$
- (b)  $\frac{\mu_0}{2\pi} \frac{ni}{r}$
- (c)  $\frac{\mu_0 ni}{4r}$
- (d)  $\mu_0 ni$ .

Q5. The wing span of an aeroplane is 20 metres. It is flying in a field, where the vertical component of magnetic field of earth is  $5 \times 10^{-5}$  tesla, with velocity 360 km/h. The potential difference produced between the blades will be

- (a) 0.10 V
- (b) 0.15 V

- (c) 0.20 V                      (d) 0.30 V

Q6. If a current  $I$  given by  $I_0 \sin\left(\omega t - \frac{\pi}{2}\right)$  flows in an ac circuit across which an ac potential of  $E = E_0 \sin \omega t$  has been applied, then the power consumption  $P$  in the circuit will be

- (a)  $P = \frac{E_0 I_0}{\sqrt{2}}$                       (b)  $P = \sqrt{2} E_0 I_0$   
 (c)  $P = \frac{E_0 I_0}{2}$                       (d)  $P = 0$ .

Q7. Which of the following is used to investigate the structure of crystal?

- a. Gamma Rays      b. Infrared Rays  
 c. X-Rays              d. Cosmic Rays.

Q8. Match List I with List II and select the correct answer using the codes given below the lists:

**List I**

(Position of the object)

(I) An object is placed at focus before a convex mirror

(II) An object is placed at centre of curvature before a concave mirror

(III) An object is placed at focus before a concave mirror

(IV) An object is placed at centre of curvature before a convex mirror

**List II**

(Magnification)

(A) Magnification is  $-\infty$

(B) Magnification is 0.5

(C) Magnification is +1

(D) Magnification is  $-1$

(E) Magnification is 0.33

Codes:

(a) I-B, II-D, III-A, IV-E

(b) I-A, II-D, III-C, IV-B

(c) I-C, II-B, III-A, IV-E

(d) I-B, II-E, III-D, IV-C

Q9. Two coherent sources of light can be obtained by

- (a) Two different lamps  
 (b) Two different lamps but of the same power  
 (c) Two different lamps of same power and having the same colour

(d) None of the above

Q10. A particle which has zero rest mass and non-zero energy and momentum must travel with a speed

- (a) Equal to  $c$ , the speed of light in vacuum
- (b) Greater than  $c$
- (c) Less than  $c$
- (d) Tending to infinity

Q11. In the  $n^{\text{th}}$  orbit, the energy of an electron  $E_n = -\frac{13.6}{n^2} \text{ eV}$  for hydrogen atom. The energy required to take the electron from first orbit to second orbit will be

- (a) 10.2 eV      (b) 12.1 eV
- (c) 13.6 eV      (d) 3.4 eV

Q12. Wires  $P$  and  $Q$  have the same resistance at ordinary (room) temperature. When heated, resistance of  $P$  increases and that of  $Q$  decreases. We conclude that

- (a)  $P$  and  $Q$  are conductors of different materials
- (b)  $P$  is  $N$ -type semiconductor and  $Q$  is  $P$ -type semiconductor
- (c)  $P$  is semiconductor and  $Q$  is conductor
- (d)  $P$  is conductor and  $Q$  is semiconductor

For Questions 13 to 16, two statements are given –one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

- A. If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- B. If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- C. If Assertion is true but Reason is false.
- D. If both Assertion and Reason are false.
- E.

Q13. **Assertion:** Electromagnetic waves are transverse in nature.

**Reason:** The electric and magnetic fields in electromagnetic waves are perpendicular to each other and the direction of propagation.

Q14. **Assertion:** When a tiny circular obstacle is placed in the path of light from some distance, a bright spot is seen at the centre of shadow of the obstacle.

**Reason:** Destructive interference occurs at the centre of the shadow.

Q15. **Assertion:** Density of all the nuclei is same.

**Reason:** Radius of nucleus is directly proportional to the cube root of mass number.

Q16. **Assertion:** The electron in the hydrogen atom passes from energy level  $n = 4$  to the  $n = 1$  level. The maximum and minimum number of photons that can be emitted are six and one respectively.

**Reason:** The photons are emitted when electron make a transition from the higher energy state to the lower energy state.

### [SECTION – B]

(05x2=10 marks)

Q17. Estimate the average drift speed of conduction electrons in a copper wire of cross-sectional area  $1.0 \times 10^{-7} \text{ m}^2$  carrying a current of 1.5 A. Assume that the density of conduction electrons to be  $9 \times 10^{28} \text{ m}^{-3}$ .

OR

Plot a graph showing temperature dependence of resistivity for a typical semiconductor. How is this behaviour explained?

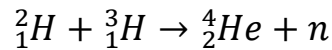
Q18. A bar magnet of dipole moment  $3 \text{ A-m}^2$  rests with its centre on a frictionless pivot. A force  $F$  is applied at right angles to the axis of the magnet, 10 cm from the pivot. It is observed that an external magnetic field of 0.25 T is required to hold the magnet in equilibrium at an angle of  $30^\circ$  with the field. Calculate the value of  $F$ . How will the equilibrium be affected, if  $F$  is withdrawn?

Q19. Two monochromatic waves emanating from two coherent sources have the displacements represented by,  $y_1 = a \cos \omega t$  and  $y_2 = a \cos(\omega t + \phi)$ , where,  $\phi$  is the phase difference between the two waves. Show that the resultant intensity at a point due to their superposition is given by  $I = 4I_0 \cos^2 \frac{\phi}{2}$ , where  $I_0 = a^2$ .

Q20. Photoelectric emission occurs when a surface is irradiated with the radiation of frequency (i)  $\nu_1$  and (ii)  $\nu_2$ . The maximum kinetic energy of the

electrons emitted in the two cases are  $K$  and  $2K$  respectively. Obtain the expression for the threshold frequency for the surface.

Q21. Calculate the energy release in MeV in the deuterium-tritium fusion reaction.



Using the data

$$m({}^2_1H) = 2.014102 \, u,$$

$$m({}^3_1H) = 3.016049 \, u,$$

$$m({}^4_2He) = 4.002603 \, u,$$

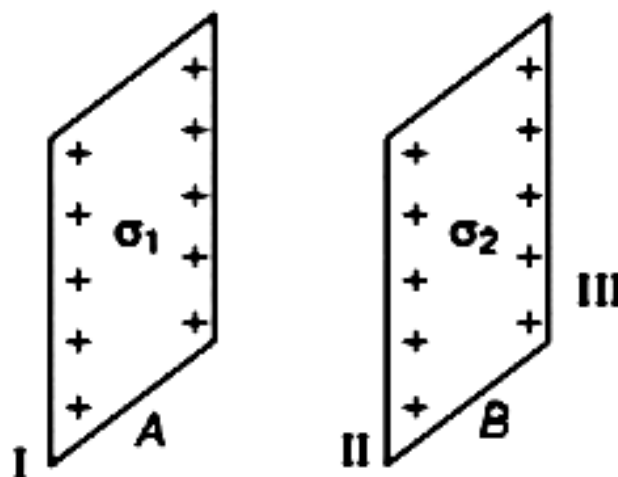
$$m_n = 1.008665 \, u$$

$$1u = \frac{931.5}{c^2} \, \text{MeV}.$$

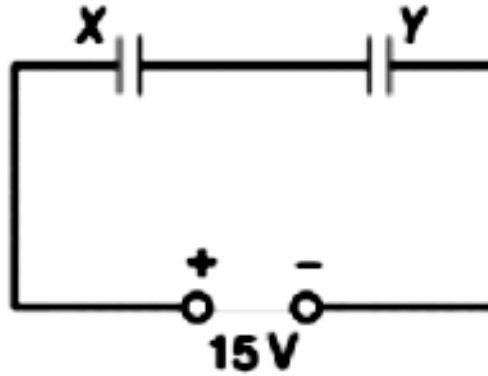
### [SECTION – C]

(07x3=21 marks)

Q22. Two infinitely large plane thin parallel sheets having surface charge densities  $\sigma_1$ , and  $\sigma_2$  ( $\sigma_1 > \sigma_2$ ) are shown in the figure. Write the magnitudes and directions of the net fields in the regions marked I, II and III.



Q23. Two parallel plate capacitors X and Y have the same area of plates and same separation between them, X has air between the plates while Y contains a dielectric medium of  $\epsilon_r = 4$ .



- (i) Calculate the capacitance of each capacitor if equivalent capacitance of the combination is  $4\mu\text{F}$ .
- (ii) Calculate the potential difference between the plates of X and Y.
- (iii) Estimate the ratio of electrostatic energy stored in X and Y.

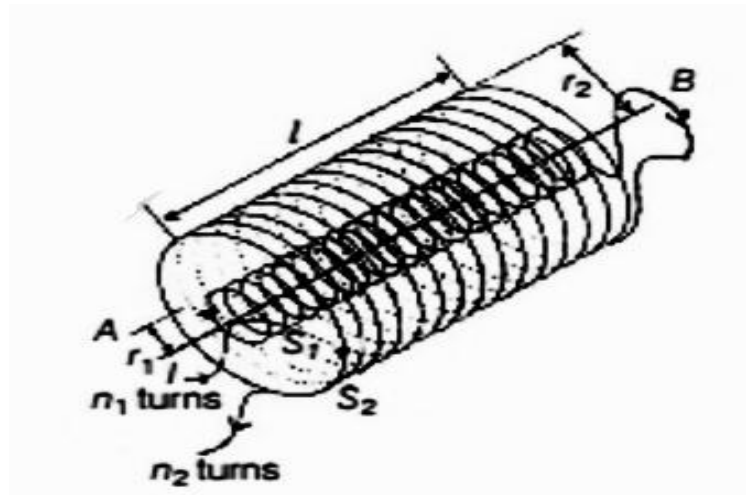
OR

A parallel plate capacitor is charged by a battery. After sometime, the battery is disconnected and a dielectric slab with its thickness equal to the plate separation is inserted between the plates. How will (i) the capacitances of the capacitor (ii) potential difference between the plates (iii) the energy stored in the capacitors be affected? Justify your answer in each case.

Q24. (i) State Ampere's circuital law expressing it in the integral form.

(ii) Two long co-axial insulated solenoids  $S_1$  and  $S_2$  of equal length are wound one over the other as shown in the figure. A steady current  $I$  flows through the inner solenoid  $S_1$ , to the other end B which is connected to the outer solenoid  $S_2$  through which the same current  $I$  flows in the opposite direction so, as to come out at end A. If  $n_1$ , and  $n_2$  are the number of turns per unit length, find the magnitude and direction of the net magnetic field at a point

- (a) inside on the axis and
- (b) outside the combined system.



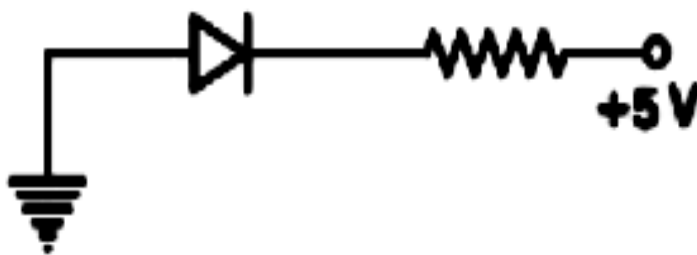
Q25. A convex lens made up of a glass of refractive index 1.5 is dipped in

- (i) a medium of refractive index 1.65
- (ii) a medium of refractive index 1.33.

- (a) Will it behave as a converging lens or a diverging lens in the two cases?
- (b) How will its focal length change in the two media?

Q26. Two objects P and Q when placed at different positions in front of a concave mirror of focal length 20 cm, form real images of equal size. Size of object P is three times size of object Q. If the distance of P is 50 cm from the mirror, find the distance of Q from the mirror.

Q27. (i) In the following diagram, is the junction diode forward biased or reverse biased?



(ii) Draw the circuit diagram of a full wave rectifier and state how it works?

Q28. Draw the energy band diagrams of (i) n-type and (ii) p-type semiconductor at temperature,  $T > 0$  K. In the case n-type Si semiconductor, the donor energy level is slightly below the bottom of conduction band whereas in p-type semiconductor, the acceptor energy level is slightly above the top of the valence band. Explain, what role do these energy levels play in conduction and valence bands.



**[SECTION D]****(02x4=08 marks)**

**Q29. Case Based Questions** Directions (Q.29) This question is case study-based question. Attempt any 4 sub-parts from this question. Each question carries 1 mark.

An electron with a speed  $v_o \ll c$  moves in a circle of radius  $r_o$  in a uniform magnetic field. This electron can traverse a circular path as magnetic field is perpendicular to the velocity of the electron. The time required for one revolution of the electron is  $T_o$ . The speed of the electron is now increase to  $4 v_o$

(i) The radius of the circle will change to

- (a)  $4r_o$                       (b)  $2r_o$                       (c)  $\frac{r_o}{4}$                       (d)  $\frac{r_o}{2}$ .

(ii) The time required for one revolution of the electron will change to

- (a)  $4T_o$                       (b)  $T_o$                       (c)  $2T_o$                       (d)  $\frac{T_o}{2}$ .

(iii) A charged particle is projected in a magnetic field  $B = (2\hat{i} + 4\hat{j}) \times 10^2 T$ . The acceleration of the particle is found to be  $a = (x\hat{i} + 4\hat{j})ms^{-2}$ . Find the value of x.

- (a)  $8 ms^{-2}$                       (b)  $-8 ms^{-2}$   
(c)  $-4 ms^{-2}$                       (d)  $4 ms^{-2}$

(iv) If the given electron has a velocity not perpendicular to B, then the trajectory of the electron is

- (a) straight line                      (b) circular  
(c) helical                      (d) Both (a) and (c).

(v) If this electron of charge (e) is moving parallel to uniform magnetic field (B) with constant velocity v. The force acting on the electron is

- (a) Bev                      (b) 2 Bev                      (c)  $\frac{Bev}{2}$                       (d) zero.

**Q30. Case Based Questions** Directions (Q.30) This question is case study-based question. Attempt any 4 sub-parts from this question. Each question carries 1 mark.

Matter cannot exist both as a particle and as a wave simultaneously. At a particular instant of time, it is either the one or the other aspect, i.e. the two aspects are complementary to each other. According to de-Broglie, a wave

is associated with moving material particle which controls the particle in every respect. The wave associated with moving material particle is called matter wave or de-Broglie wave whose wavelength is called de-Broglie wavelength.

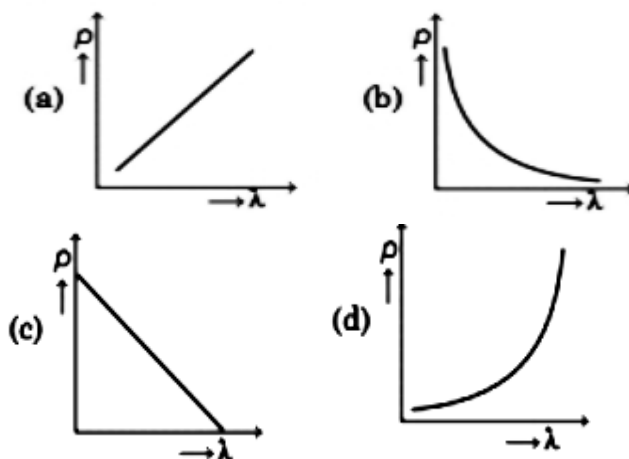
(i) The de-Broglie wave of a moving particle does not depend on

(a) mass (b) charge (c) velocity (d) momentum.

(ii) The de-Broglie wavelength of a particle of KE,  $K$  is  $\lambda$ . What will be the wavelength of the particle, if its kinetic energy is  $\frac{K}{9}$ .

(a)  $\lambda$  (b)  $2\lambda$  (c)  $3\lambda$  (d)  $4\lambda$ .

(iii) Which of the following figures represent the variation of particle momentum and the associated de-Broglie wavelength?



(iv) de-Broglie wavelength associated with an electron, accelerating through a potential difference of 100 V lies in the region of

(a) Gamma rays  
(b) X-rays  
(c) Ultraviolet rays  
(d) Visible region.

(v) A proton and an  $\alpha$ -particle are accelerated through the same potential difference. The ratio of de-Broglie wavelength  $\lambda_p$  to that of  $\lambda_\alpha$  is

(a)  $\sqrt{2}:1$   
(b)  $\sqrt{4}:1$   
(c)  $\sqrt{6}:1$   
(d)  $\sqrt{8}:1$ .

**[SECTION E] (03X5=15)**

Q31. (i) Derive an expression for drift velocity of electrons in a conductor. Hence, deduce Ohm's law.

(ii) A wire whose cross-sectional area is increasing linearly from its one end to the other is connected across a battery of potential difference  $V$  volt.

Which of the following quantities remain constant in the wire?

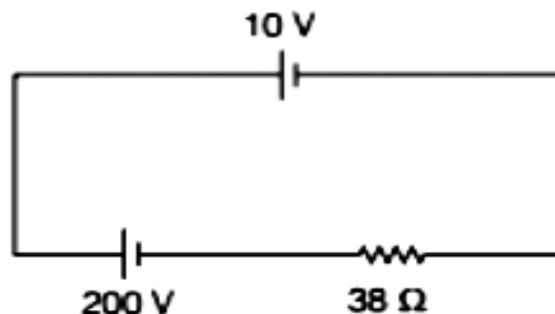
- (a) Drift speed                                      (b) Current density  
(c) Electric current                              (d) Electric field.

Justify your answer.

OR

(a) Distinguish between emf ( $\varepsilon$ ) and terminal voltage ( $V$ ) of a cell having internal resistance  $r$ . Draw a plot showing the variation of terminal voltage ( $V$ ) versus the current ( $I$ ) drawn from the cell. Using this plot show how does one can determine the internal resistance of the cell?

(b) A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance  $38 \Omega$  as shown in the figure. Find the value of current in the circuit.



**Q32.** A device  $X$  is connected across an AC source of voltage  $V = V_o \sin \omega t$ .

The current through  $X$  is given as  $I = I_o \sin \left( \omega t + \frac{\pi}{2} \right)$

- (i) Identify the device  $X$ .
- (ii) Draw graphs showing variation of voltage and current with time over one cycle of AC, for  $X$ .
- (iii) How does the reactance of the device  $X$  vary with frequency of the AC? Show this variation graphically.
- (iv) Draw the phasor diagram for the device  $X$ .

OR

(a) Draw the diagram of a device which is used to decrease high AC voltage into a low AC voltage and state its working principle. Write four sources of energy loss in this device.

(b) A small town with a demand of 1200 kW of electric power at 220 V is situated 20 km away from an electric plant generating power at 440 V. The resistance of the two-wire line carrying power is  $0.5 \Omega$  per km. The town gets the power from the line through a 4000-220 V step-down transformer at a sub-station in the town. Estimate the line power loss in the form of heat.

**Q33.** (i) Draw a labelled ray diagram to obtain the real image formed by an astronomical telescope in normal adjustment position. Define its magnifying power.

(ii) You are given three lenses of power 0.5 D, 4 D and 10 D to design a telescope.

(a) Which lenses should be used as objective and eyepiece? Justify your answer.

(b) Why is the aperture of the objective preferred to be large?

OR

- a. Draw a ray diagram to show the working of a compound microscope. Deduce an expression for the total magnification when the final image is formed at the near point.
- b. In a compound microscope, an object is placed at 1.5 cm from the objective of focal length 1.25 cm. If the eyepiece has a focal length of 5 cm and the final image is formed at the near point. Estimate the magnifying power of the microscope.